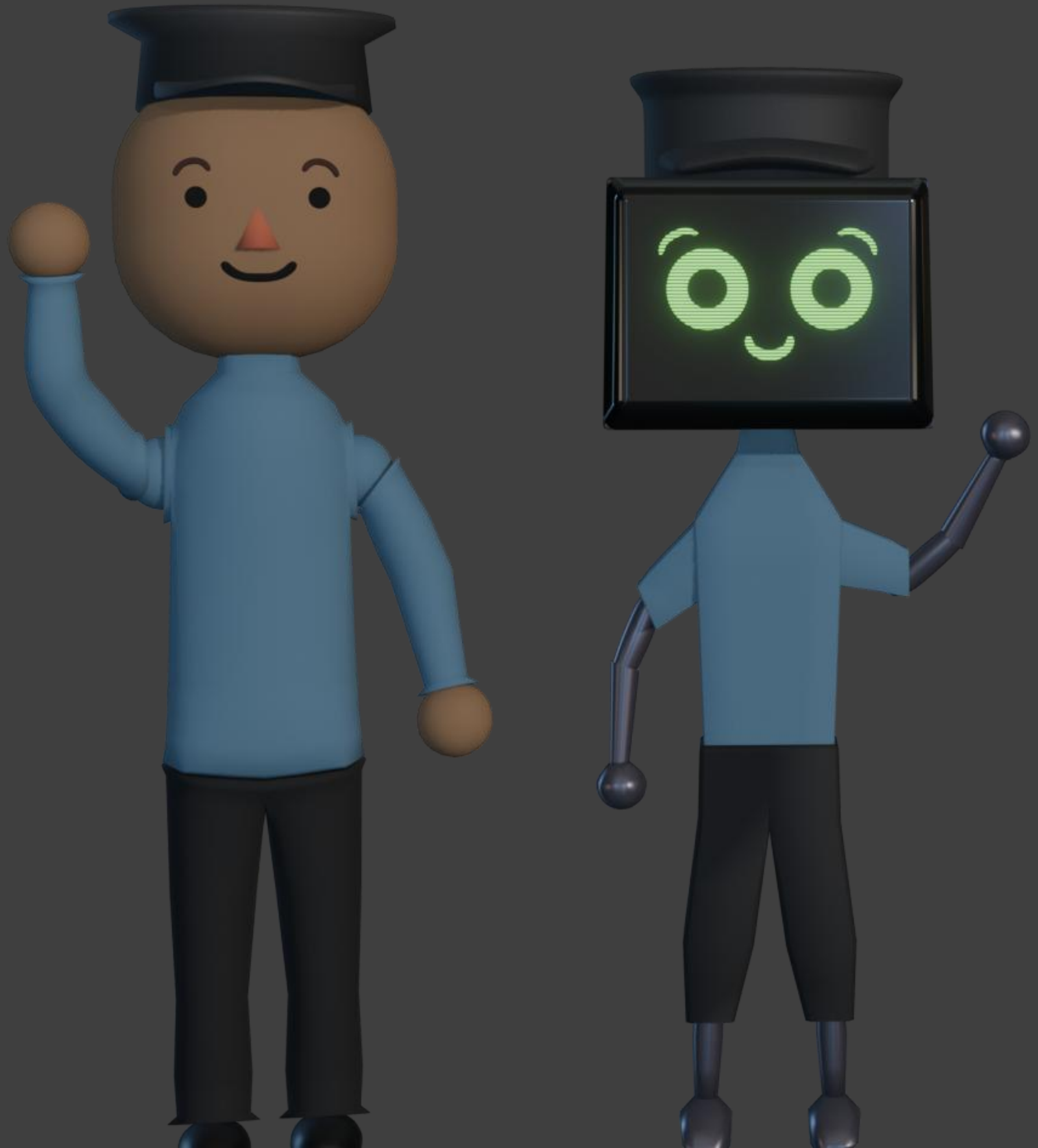

A Human-Security Robot Interaction Literature Review

https://doi.org/10.1145/3700888

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Presentation

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Why Security Robots?

- Cost-effective policing
- Can handle dangerous situations

Why not Security Robots?

- Challenges with human interaction
  - Civilians and security personnel
- A lot more unsaid

N.Y.P.D. Robot Dog's Run Is Cut Short After Fierce Backlash

The Police Department will return the device earlier than planned after critics seized on it as a dystopian example of overly aggressive policing.

Share full article

The Police Department used a robotic dog like this one from Boston Dynamics. The machine, which the police named Digidog, became a source of heated debate. Josh Reynolds/Associated Press

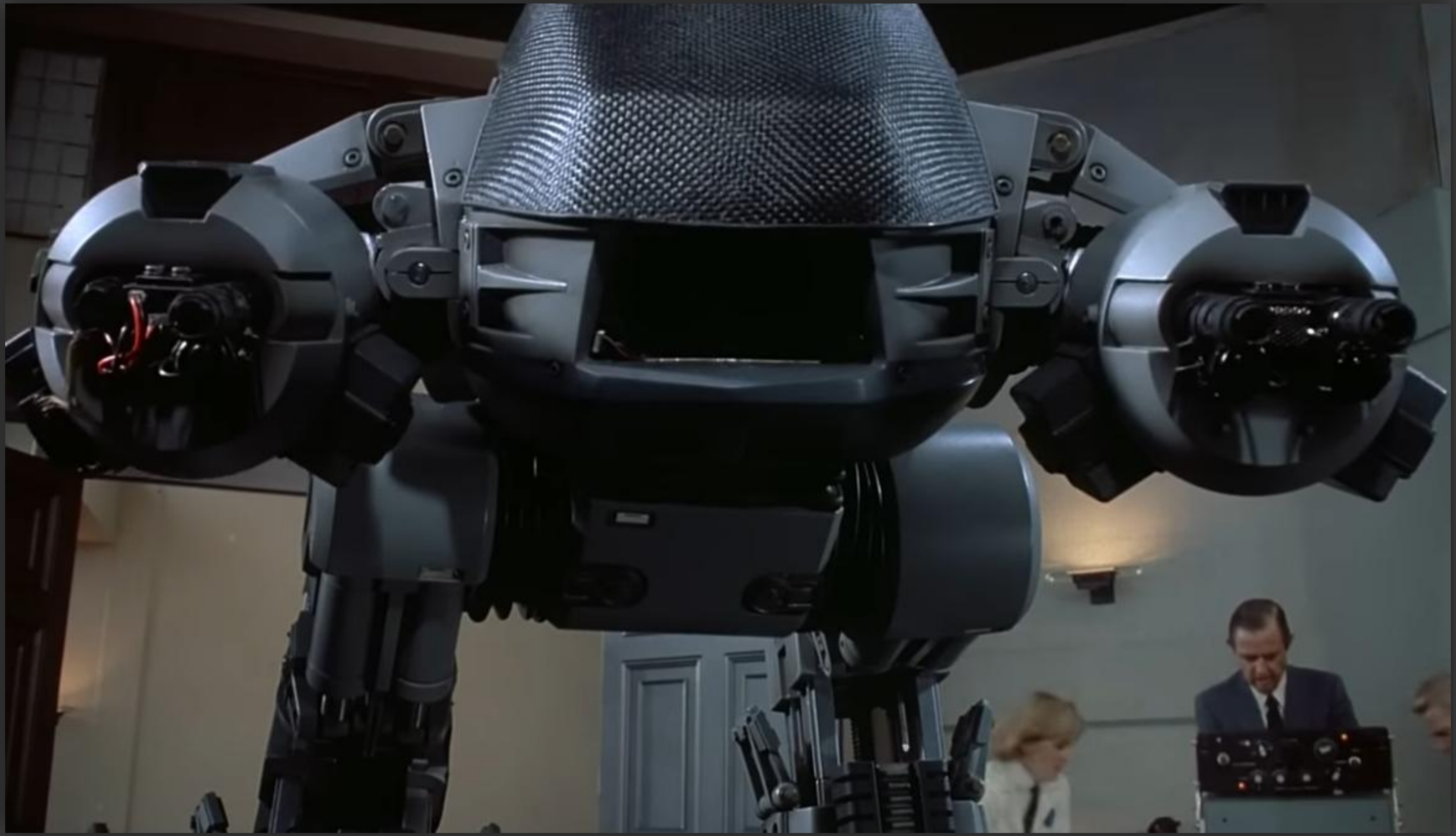
Mihir Zaveri, New York Times, 2021

	Robots (4.1.1)	Security Robots (4.1.2)
Characteristics	Artificial intelligence, some level of autonomy	Facial recognition, cameras, detection, recording, sensing, weapons (e.g., taser, gun), some level of autonomous decision-making
Examples	Autonomous vacuum cleaners, voice assistants, self-checkout machines, food delivery robots	Passive surveillance and data collection, theft prevention, identifying wanted criminals, calling law enforcement to scene or intervening itself
Influential sources on perceptions	Instagram videos of food delivery robots, news coverage on the work of robotics company Boston Dynamics, Rosie the robot housekeeper in The Jetsons cartoon	Dystopian films such as RoboCop, reports of China using facial recognition to surveil citizens, controversy in Europe over use of facial recognition for traffic enforcement

	Perceived Benefits	Perceived Risks and Concerns
Presence in the community	Deterrent to crime (4.2.1)	Potential for deepening social inequity (4.3.1)
Robot action and human agenda	Prevent or respond to events (4.2.2)	Unease about unclear source of control (4.3.2); Malfunction/hacking threatening physical safety (4.3.3)
Implications for specific populations	Lower threat of violence against women (4.2.3)	Likelihood of biased judgements and discrimination (4.3.4)

Summary of interviews concerning security robots in 2020. n=17!

Gabriela Marcu et al. "Would I Feel More Secure With a Robot?": Understanding Perceptions of Security Robots in Public Spaces



ED-209 is a self-sufficient law enforcement robot.

Literature Review

- 2.1 Definition of Security Robots
- 2.2 Search Process
- 2.3 Screening Procedure

“In this study, ‘security robots’ refers to robots deployed to prevent unwanted activities through their presence, surveillance, and ability to notify authorities of unauthorized personnel or actions.”

“This broad definition acknowledges that any robot can be a security robot based on its actual use regardless of its intended purpose.”

p21:1

2.1 Definition of Security Robots

Can a vacuum be a security robot?

If it surveils the premises and phones home, sure, I guess.

2.1 Definition of Security Robots

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If it surveils the premises and phones home, sure, I guess.

2.1 Definition of Security Robots

- Design-Specific Security Robots (DSSRs)
  - Explicitly designed for security (ie. patrolling and monitoring areas)
- Non-Design-Specific Security Robots (DSSRs)
  - Ex. Customer service may monitor security-related data
  - Some military robots
    - Filtered based on operator, purpose, environment, protagonist/antagonist model
- Excluded robots
  - Rescue robots, industrial robots, medical assistant robots

2.1 Definition of Security Robots

RAMSEE (left) and Knightscope K5 (right) are
design-specific security robots (DSSRs)

2.2 Search Process

Search Engines

- Google Scholar
- ACM Digital Library
- IEEE Explore
- Scopus

Search Terms

- Security
- Peacekeeping
- Guard
- Police
- Military
- Safety
- Patrol
- Protection
- Robot(s) [&]

Details

- 23 August 2023
- 4,116 results after duplicates removed
- **47 studies after filtering**

2.3 Screening Procedure

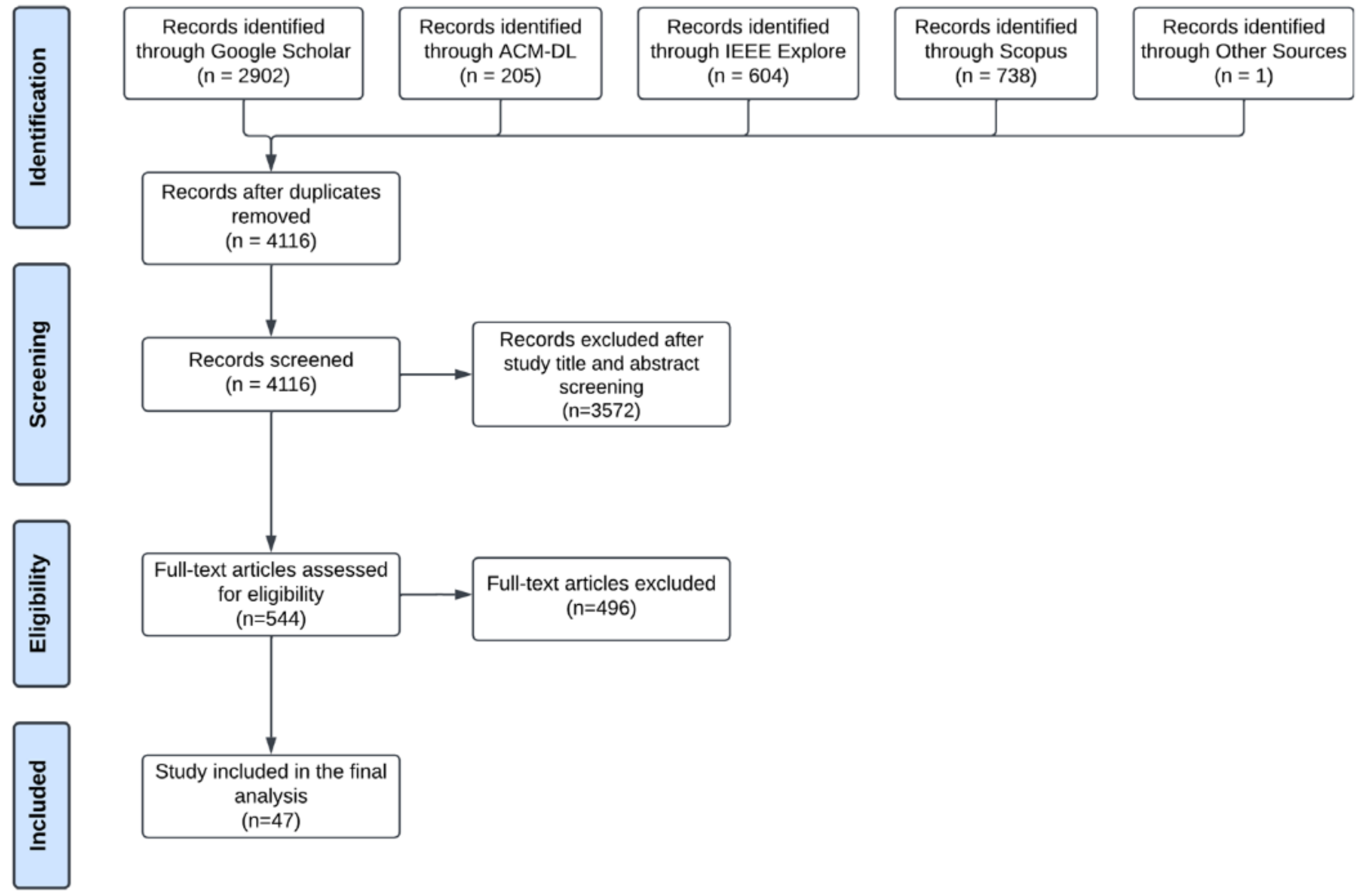
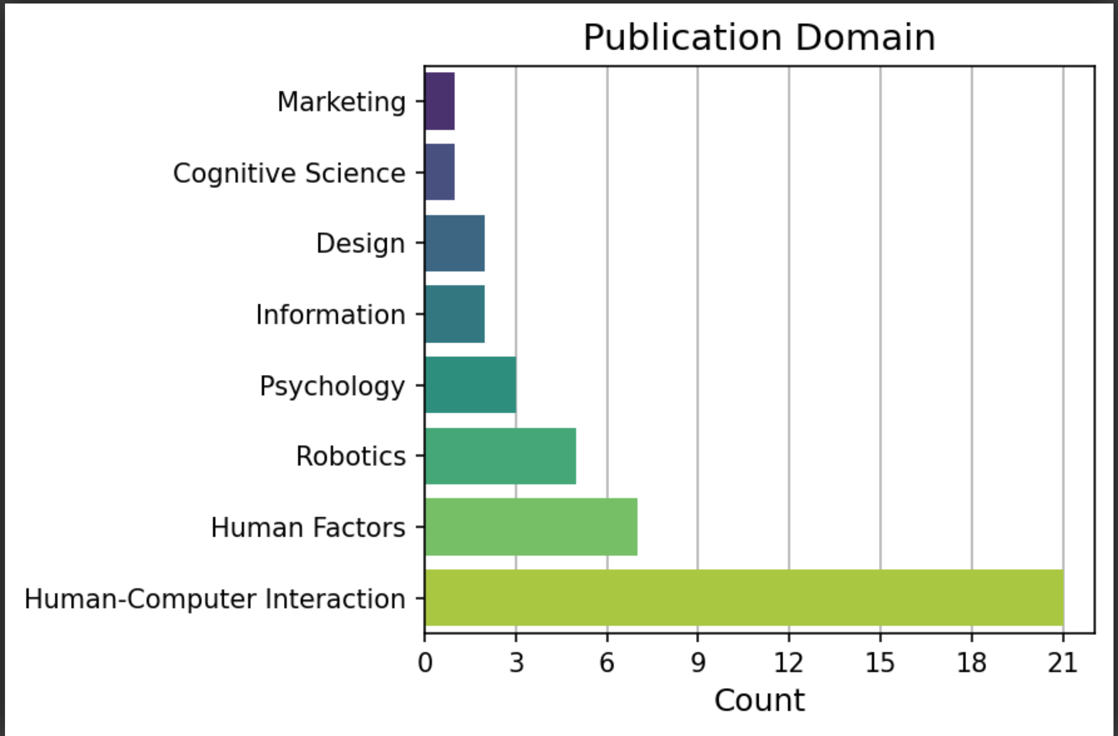


Fig. 1. PRISMA flow diagram of the literature review process.

Review Results

- 3.1 Publication Outlets
- 3.2 Sample Data
- 3.3 Interaction Methods, Application Domains, and Tasks
- 3.4 Outcomes
- 3.5 Research Thrust Areas
- 3.6 Findings

3.1 Publication Outlets



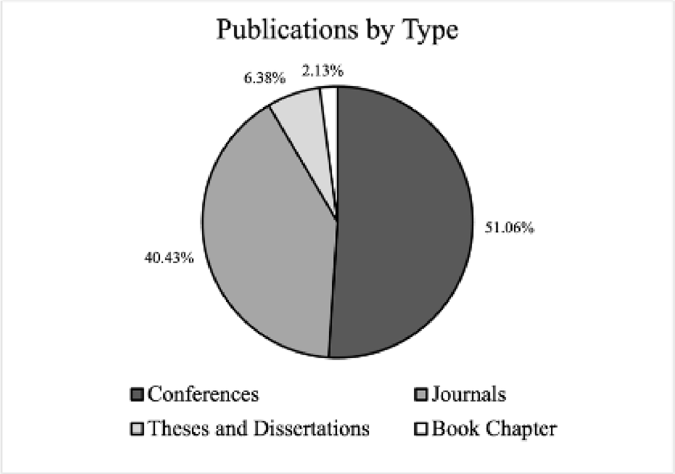


Fig. 2. Publications by type.

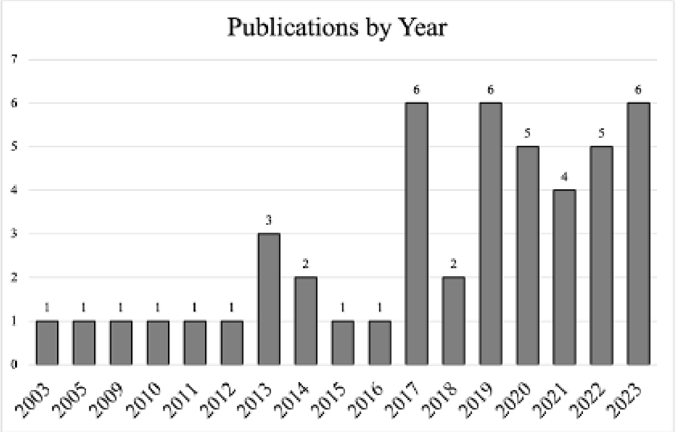


Fig. 3. Publications by year.

3.2 Sample Data

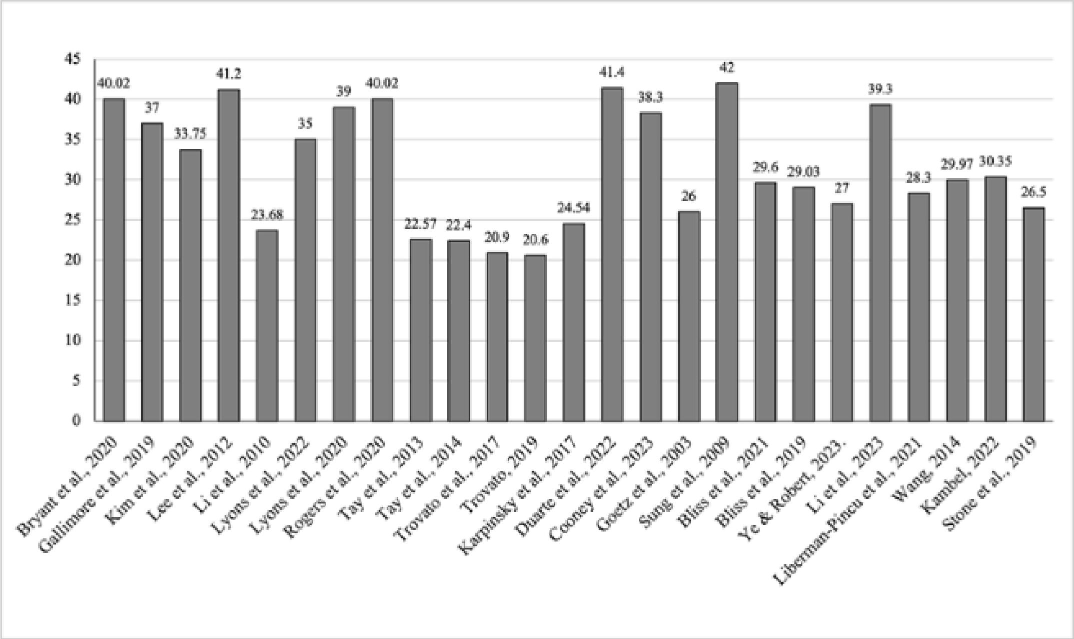


Fig. 5. Average participant age by study.

The overall gender distributions was well balanced (F:M=48.49:51.51]) but individual studies varied significantly and 14 studies did not specify

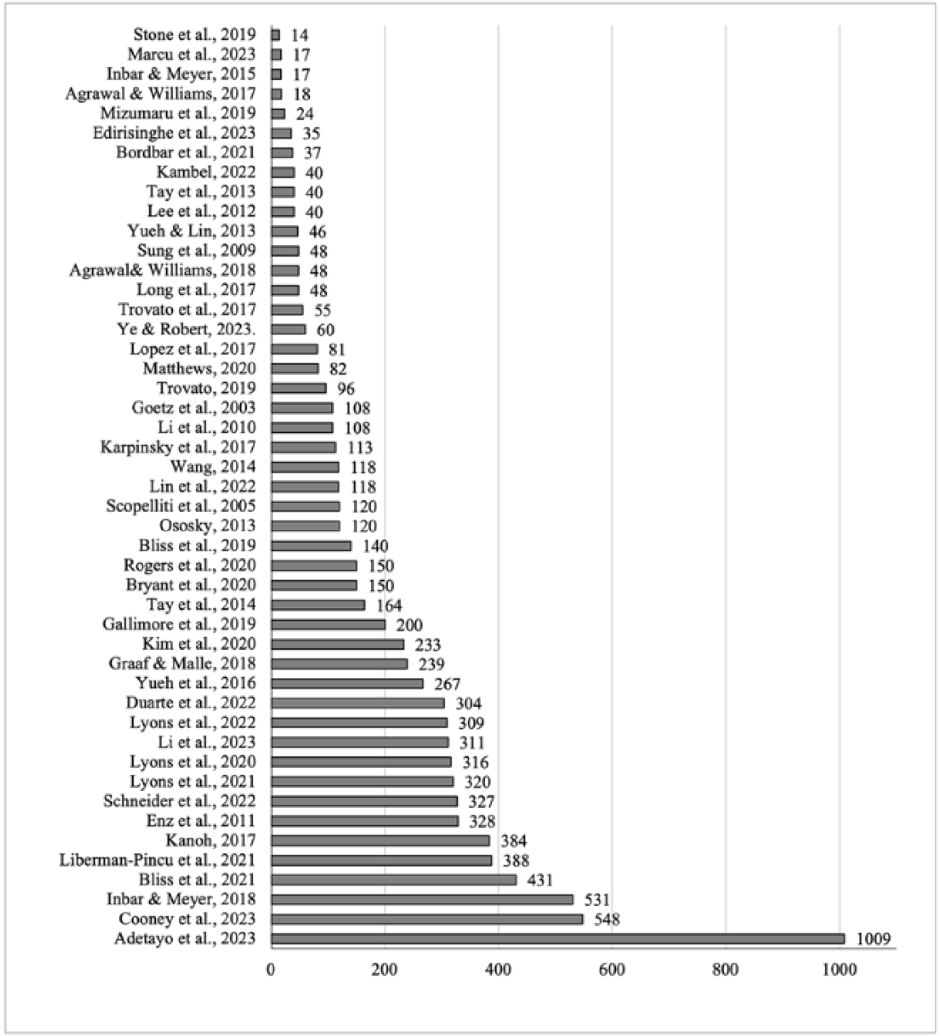
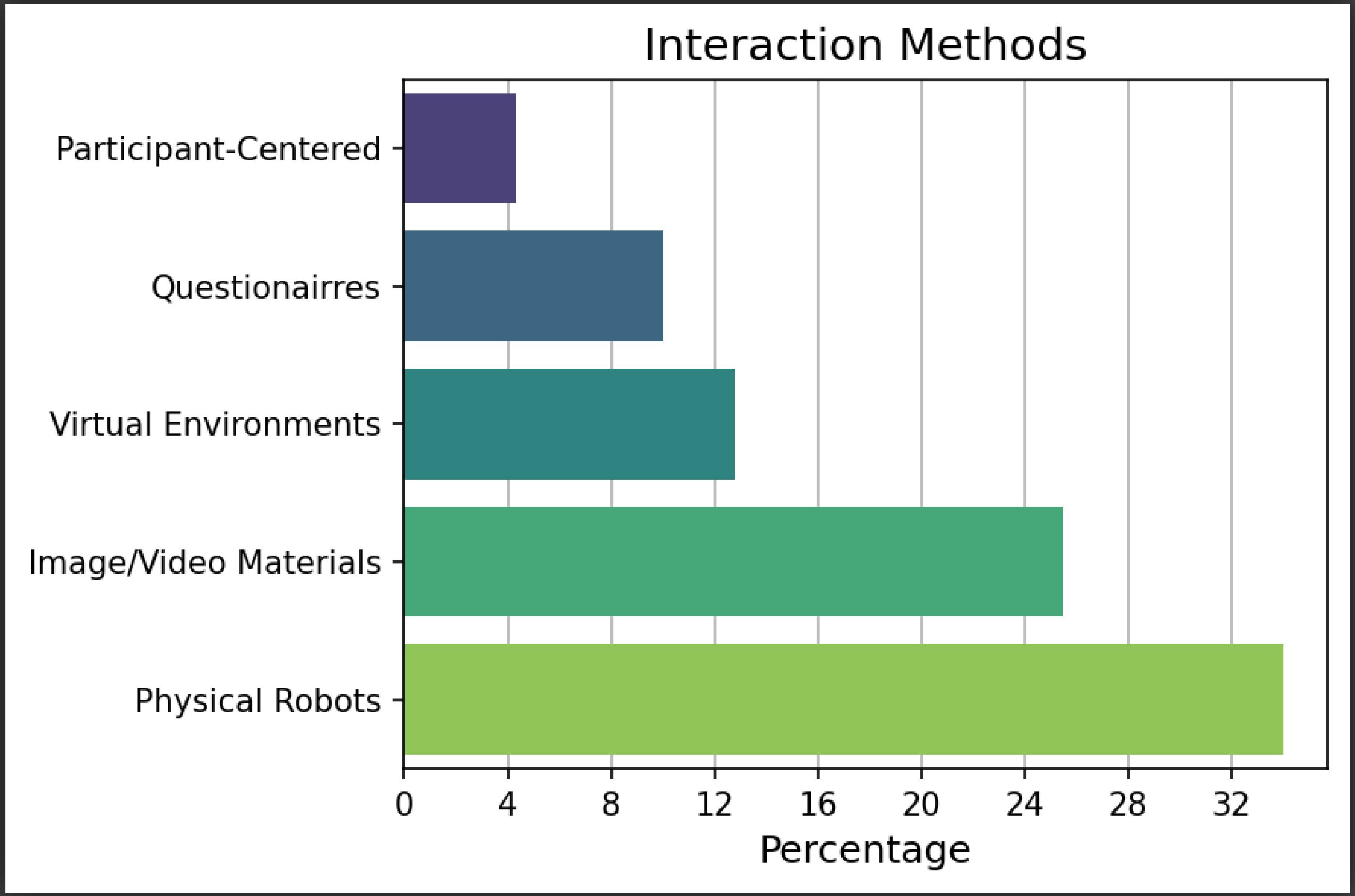


Fig. 4. Participants by study.

3.3 Interaction Methods, Application Domains, and Tasks



3.3 Interaction Methods, Application Domains, and Tasks

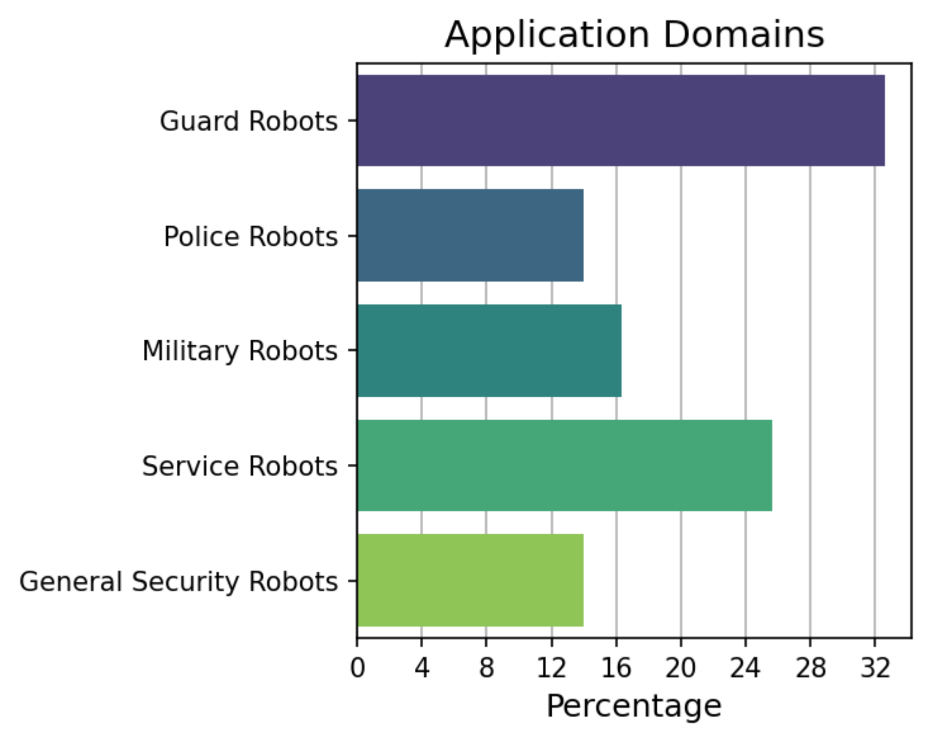


Table 1. Application Domains

Application domains	Description
Guard robot	Deployed by private agencies in settings such as campuses, airports, and markets to provide guard and protection.
Police robot	Deployed by official police departments to fight crime and ensure safety in public areas.
Military/peace-keeping robot	Deployed in the military and army to protect the safety of civilians or soldiers.
Service robot	1. Public service robot: Deployed by agencies in hotels or markets to provide comprehensive services, including protecting the safety of people by reminding them to wear masks and patrolling to monitor surroundings. 2. Private/home service robot: Deployed by individuals or families for house security tasks to protect the safety of family members or individuals.
General security robot	Deployed to perform various security tasks without specific application scenarios or potential employers.

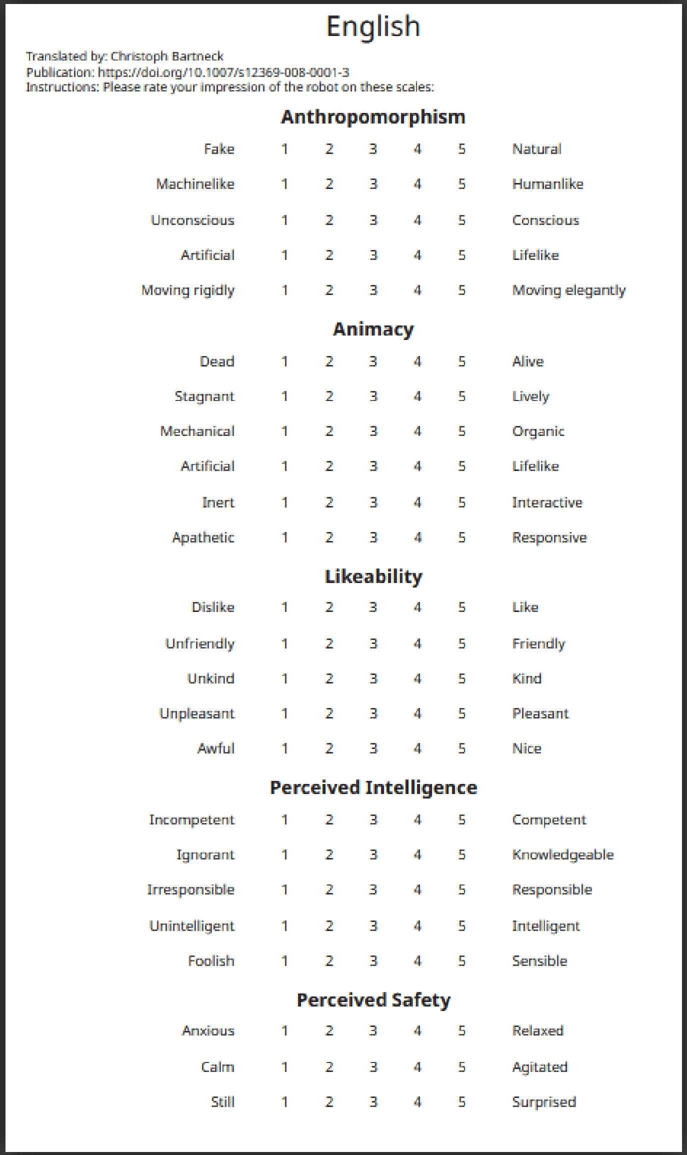
3.3 Interaction Methods, Application Domains, and Tasks

Table 2. Security Tasks

Task	Definition	Number	Paper
Access control task	Monitor and control access to a specific area.	13 (41%)	[4], [5], [34], [40], [41], [50], [54], [59–62], [93], [94]
Integrated security tasks	Simultaneous execution of multiple security functions.	7 (22%)	[3], [25], [56], [70], [88], [89], [108]
Military security task	Assist military units and provide security.	5 (16%)	[11], [12], [48], [57], [58]
Admonish task	Admonish inappropriate behaviors.	2 (6%)	[67], [81]
Protect individual task	Protect an individual human.	2 (6%)	[16], [24]
Others	Other security tasks.	3 (9%)	[13], [46], [86]
Total		32 (100%)	

3.4 Outcomes

- 3 Outcome Categories
  1. <u>Human Performance</u>
     - Least commonly studied
  2. <u>Perception</u>
     - High level evaluation and theoretical opinion
     - Ex. Godspeed questionnaire
  3. Human Acceptance
     - <u>Behavioral Acceptance</u>
       - Actual engagement outcomes
     - <u>Perceptional Acceptance</u>
       - More specifically concerned with intention to use
       - More emphasis on factors like trust and reliability



English

Translated by: Christoph Bartneck
Publication: https://doi.org/10.1007/s12369-008-0001-3
Instructions: Please rate your impression of the robot on these scales:

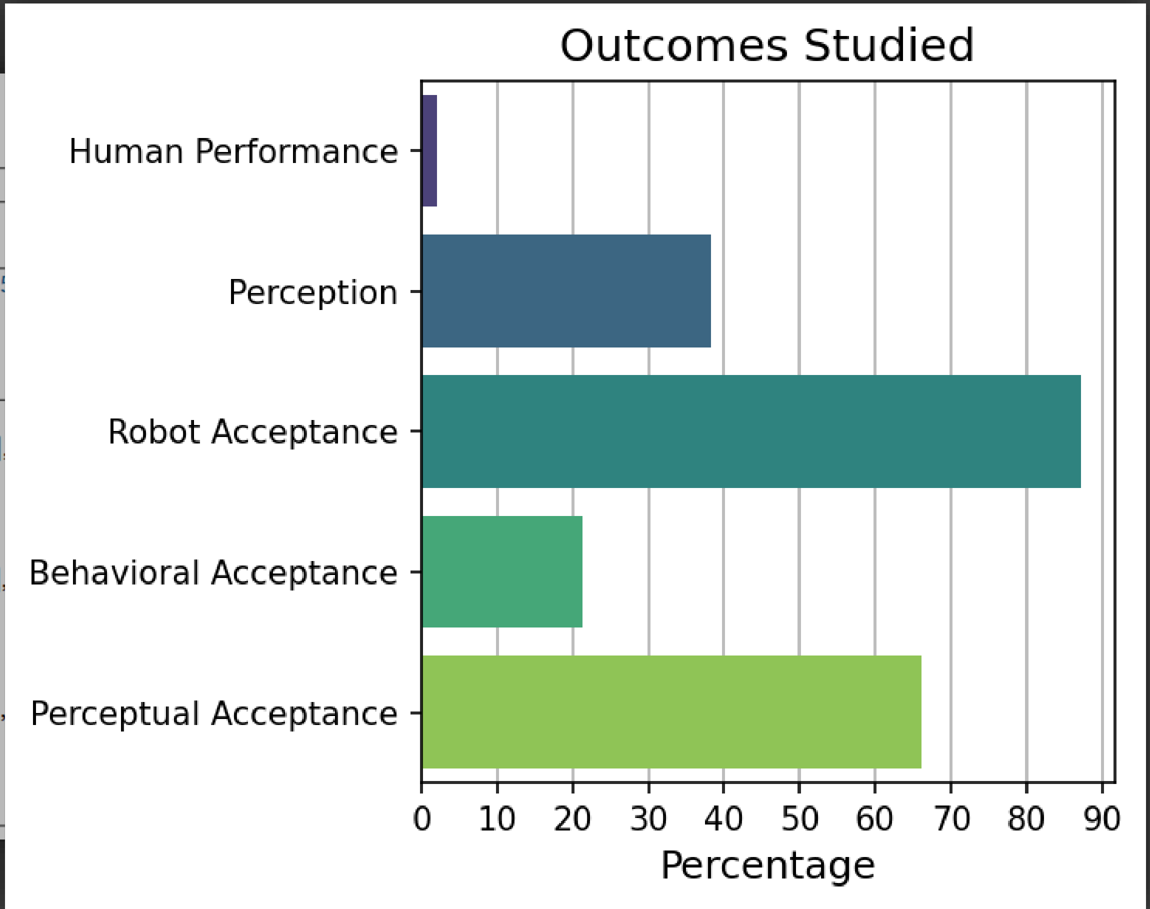
Anthropomorphism						
Fake	1	2	3	4	5	Natural
Machinelike	1	2	3	4	5	Humanlike
Unconscious	1	2	3	4	5	Conscious
Artificial	1	2	3	4	5	Lifelike
Moving rigidly	1	2	3	4	5	Moving elegantly
Animacy						
Dead	1	2	3	4	5	Alive
Stagnant	1	2	3	4	5	Lively
Mechanical	1	2	3	4	5	Organic
Artificial	1	2	3	4	5	Lifelike
Inert	1	2	3	4	5	Interactive
Apathetic	1	2	3	4	5	Responsive
Likeability						
Dislike	1	2	3	4	5	Like
Unfriendly	1	2	3	4	5	Friendly
Unkind	1	2	3	4	5	Kind
Unpleasant	1	2	3	4	5	Pleasant
Awful	1	2	3	4	5	Nice
Perceived Intelligence						
Incompetent	1	2	3	4	5	Competent
Ignorant	1	2	3	4	5	Knowledgeable
Irresponsible	1	2	3	4	5	Responsible
Unintelligent	1	2	3	4	5	Intelligent
Foolish	1	2	3	4	5	Sensible
Perceived Safety						
Anxious	1	2	3	4	5	Relaxed
Calm	1	2	3	4	5	Agitated
Still	1	2	3	4	5	Surprised

Example Godspeed questionnaire

3.4 Outcomes

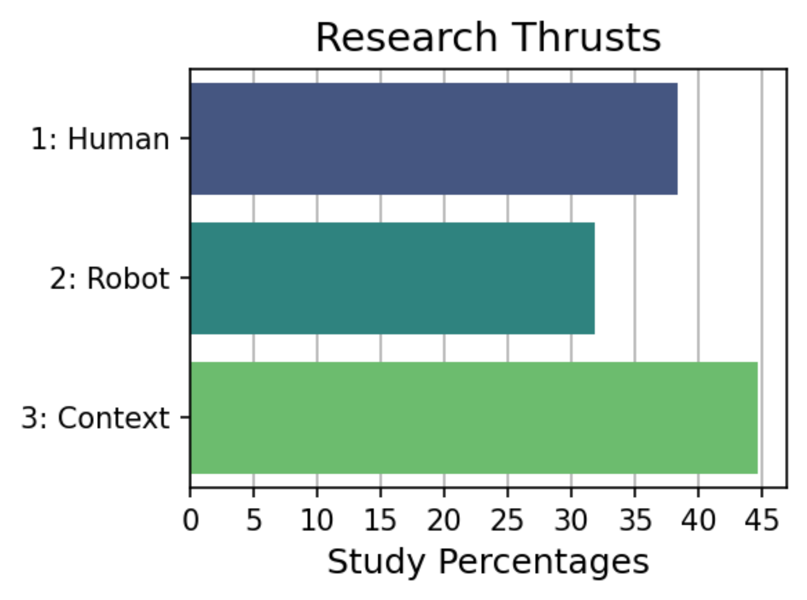
Table 3. Outcomes

Outcomes	Studies
Human Performance	Situational awareness: [86] Workload: [86]
Perception of Robots	Robot image and evaluations: [4], [5], [20], [26], [40], [41], [46], [5 Attitudes: [5], [47], [48], [53], [64], [88], [93] Godspeed: [4], [5], [20], [54], [55], [108] Interpretation of intent: [70]
Robot Acceptance	**1. Behavioral Acceptance** Engagement in interaction and response: [12], [54], [56], [59], [67], Distance from robots: [48], [93] Denial behavior: [46] **2. Perceptional Acceptance** Trust/reliance intentions: [11], [13], [14], [34], [50], [54], [57], [58], [78], [86], [88], [108] Trustworthiness: [14], [34], [50], [60], [62], [63], [70], [108] Intention to use: [34], [60–62], [78], [108], [111] Satisfaction, preference, and expectations: [3], [16], [25], [26], [35], [87–89], [103], [110] Perceived behavioral control: [88] Acceptance: [16], [24], [63]



3.5 Research Thrust Areas

Thrust Area 1: The Human in Human–Security Robot Interaction	
1-1 Gender	[14], [26], [34], [40], [41], [63], [103], [111]
1-2 Demographics (age, education, technology experience)	[26], [40], [78], [83]
1-3 Attitude	[50], [64]
1-4 Mental model	[70]
1-5 Personality	[61]
1-6 Obedience behavior	[4], [5], [81]
1-7 Guilt status	[46]
Guiding Question: How do human attributes impact human–security robot interaction?	



Thrust Area 2: The Robot in Human–Security Robot Interaction	
2-1 Robot gender	[14], [78], [88], [89]
2-2 Robot's non-physical design (stated social intent, autonomy, personality, threat analysis type, interrogation approach)	[46], [57], [60], [62–64], [88]
2-3 Robot's physical design (anthropomorphism, outlook, size, voice/volume, battery life, camera and view)	[13], [35], [54–56], [108]
2-4 Robot's non-physical behavior (politeness, communication style, cognitive dissonance intervention strategy)	[40], [41], [58], [81]
2-5 Robot's physical behavior (weapon, stand/approach/attack/defense behavior)	[5], [12], [24], [67], [87]
2-6 Reliability	[62]
2-7 Robot presence	[56], [63], [86]
Guiding Question: How do robot attributes impact human–security robot interaction?	

Thrust Area 3: The Contextual Factors in Human–Security Robot Interaction	
3-1 Robot task	[48], [58], [93], [94]
3-2 Agent type	[3], [20], [24], [25], [41], [46], [59], [78], [93]
3-3 Context and usage	[57], [63], [108]
3-4 Culture	[11], [16], [47], [48], [53], [54], [58], [78], [110]
Guiding Question: How do contextual factors impact human–security robot interaction?	

3.5 Research Thrust Areas

- Mostly a review of each outcome category for each thrust
- Human performance is often omitted
- Robot acceptance is often split into behavioral and perceptional acceptance

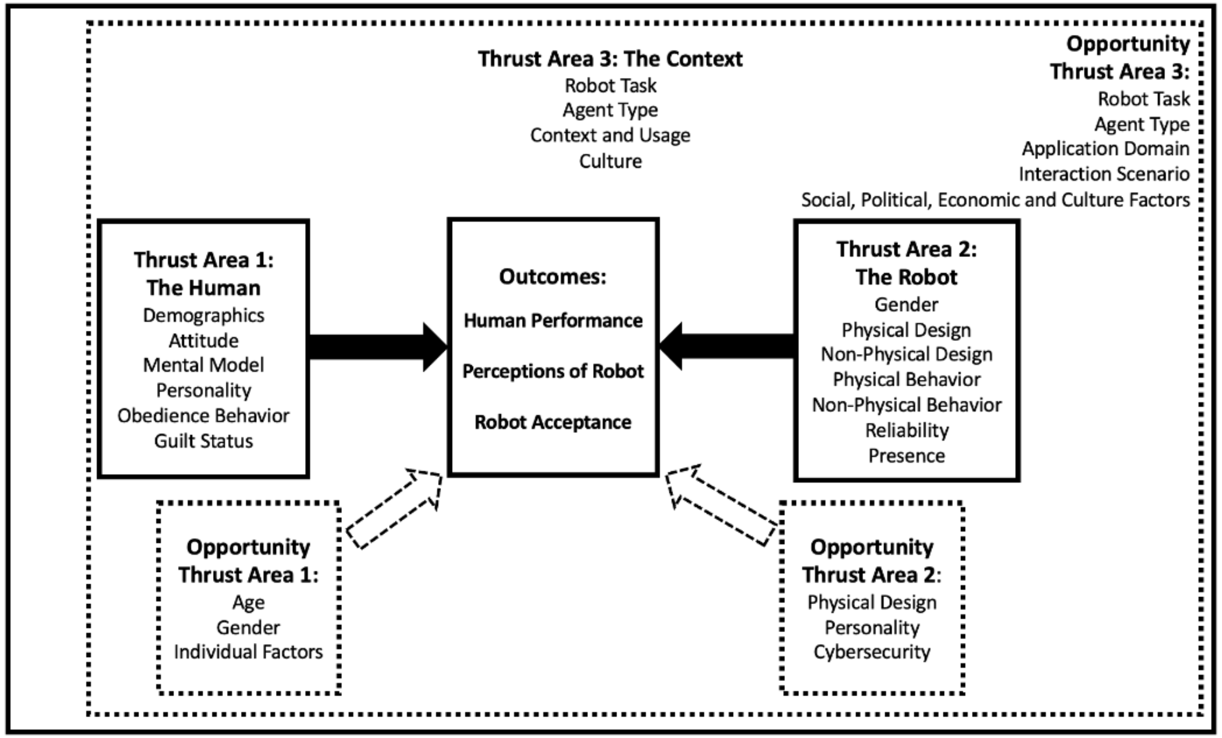


Fig. 7. Human–security robot integrative research model.

Thrust Area 1: The Human

3.6.1 Human/Perception

1. Gender not significant overall*
2. Age does not seem significant
3. Compliant individuals viewed security robots more positively safer and more positively
4. Users with security guard mental model perceive robots as better, more intelligent, and safer

3.6.1 Human/Acceptance/Behavioral

1. Cognitive dissonance (trivializing warnings, justifying ignoring robots' directives) is associated with non-compliance
2. Perceived guilt status of suspects does not reliably predict response/interactions

3.6.1 Human/Acceptance/Perceptional

1. Mixed results for impact of <u>gender</u> on acceptance with significant results indicating women show greater trust/intention to use
2. Impact of <u>age</u> is unclear with mixed results
3. <u>Familiarity with technology</u> is associated with more optimistic expectations about deploying security robots
4. <u>Personality</u>, especially agreeableness and intellect/imagination, strongly predict trust and intentions to use security robots

Six studies examined the effect of gender on perceptional acceptance.

Three found women to have higher reliance intentions on robots, but more in hospital/college type settings than public/military.

In another study, interviews indicated women perceived security robots as posing a lower risk of violence than human security.

Thrust Area 2: The Robot

3.6.2 Robot/Human Performance

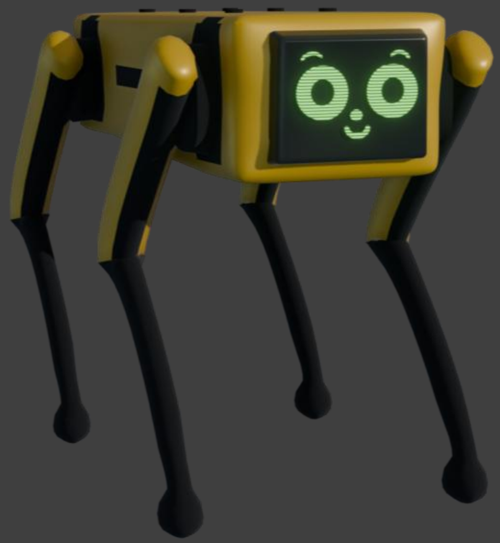
1. Use of security drones improves performance/information quality without increasing overall mental workload
2. Managing a swarm of security drones increases perceived complexity and time pressure compared to a single drone

3.6.2 Robot/Perception

1. Anthropomorphic robots are perceived as less safe than non-anthropomorphic robots, but with NO difference in likability
2. Design features like logos and flashing lights affect perception of authority/approachability, but not innovativeness, reliability, and professionalism
3. Politeness strongly influences perception; more polite security robots are viewed more favorably
4. 'Male-gendered' security robots receive better evaluations, and introverted robots are also perceived more positively

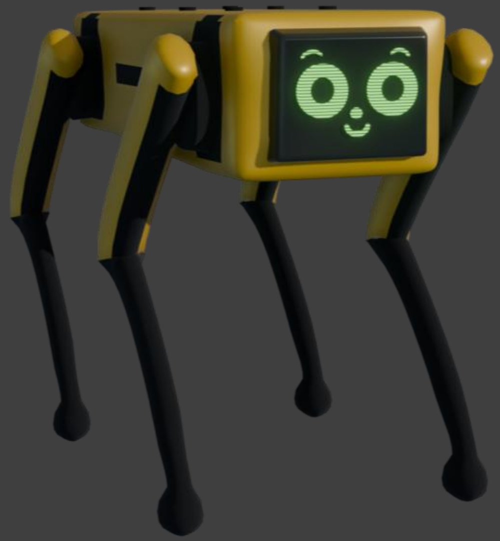
Machine-like

Machine-like



Zoomorphic

Machine-like



Zoomorphic

Anthropomorphic

SuperDroid Remote Telepresence Robot

Sony AIBO

Aldebaran Nao

A polite, introverted male robot

3.6.2 Robot/Acceptance/Behavioral

1. Visual design features (e.g. lights, logos) can affect interactions
2. The presence of a security robot can increase compliance
3. Admonishing behavior can effectively increase compliance (but appears to largely be more due to surprise than persuasion)
4. People are more inclined to comply when robots are equipped with lethal weapons than non-lethal weapons. Surprise!
5. Suspects are friendlier when interrogated with innocent-presumptive approaches rather than guilt-presumptive

Miloš, a Serbian unmanned ground vehicle, is a lethal autonomous weapon (LAW)

3.6.2 Robot/Acceptance/Perceptional

1. Impact of robot <u>gender</u> was mixed (some slight indication of viewing male robots more favorably)
2. There is preference for <u>human-like robots</u> for socially demanding tasks and machine-like otherwise
3. <u>Expressed social intent</u> influences perceptions of integrity and benevolence
4. <u>Autonomy</u> does not reduce trust outright, but raises safety and ethical concerns
5. More <u>reliable</u> security robots are trusted more. Surprise! Also, those that employ less forceful defense approaches are preferred

Thrust Area 3: Context

3.6.3 Context/Perception

1. National culture, especially regarding the use of weapons, significantly influences attitudes toward security robots
2. Mixed results for perception of human security officers vs. security robots
3. More masculine traits are attributed to security robots and more feminine traits to guidance robots
4. Little indication if or how the context of interaction itself influences perceptions of security robots

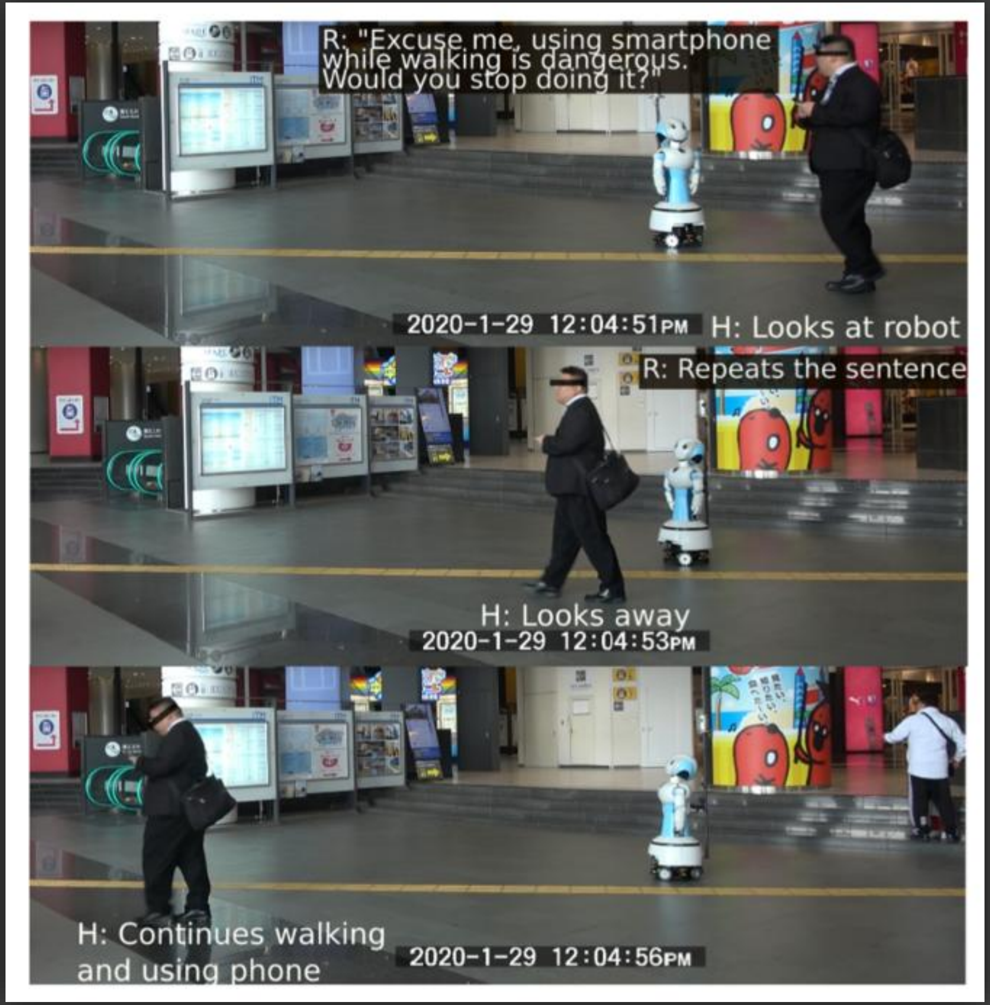
An idyllic Korean security robot

An idyllic Korean security robot

An idyllic American security robot

3.6.3 Context/Acceptance/Behavioral

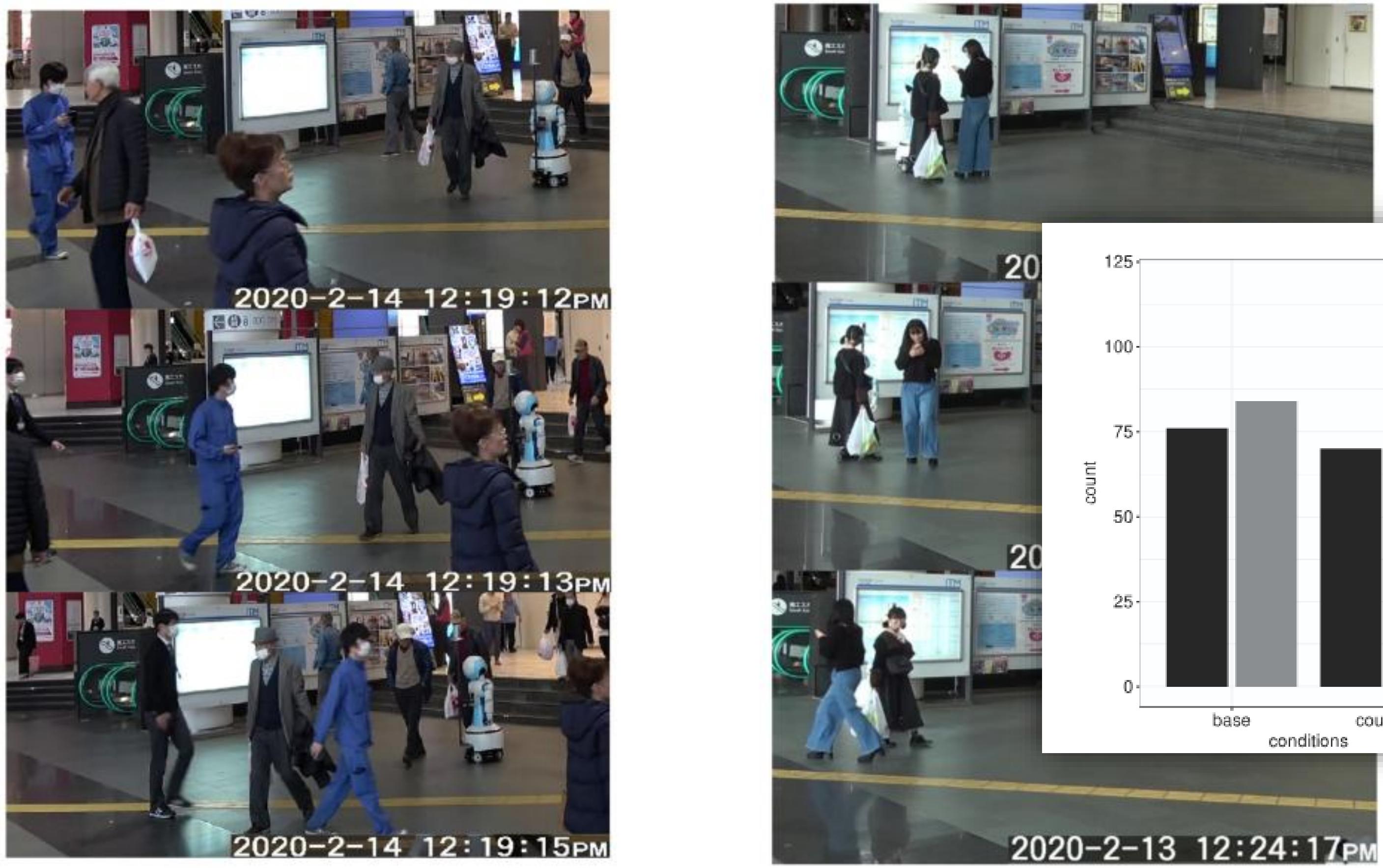
1. National culture influences human engagement and compliance
2. People are more likely to notice, interact with, and maintain distance from human security guards vs. security robots
3. Engagement varies based on task performed by robot
4. Implementing a counter-trivialization strategy in communication increases human compliance



Robovie R3

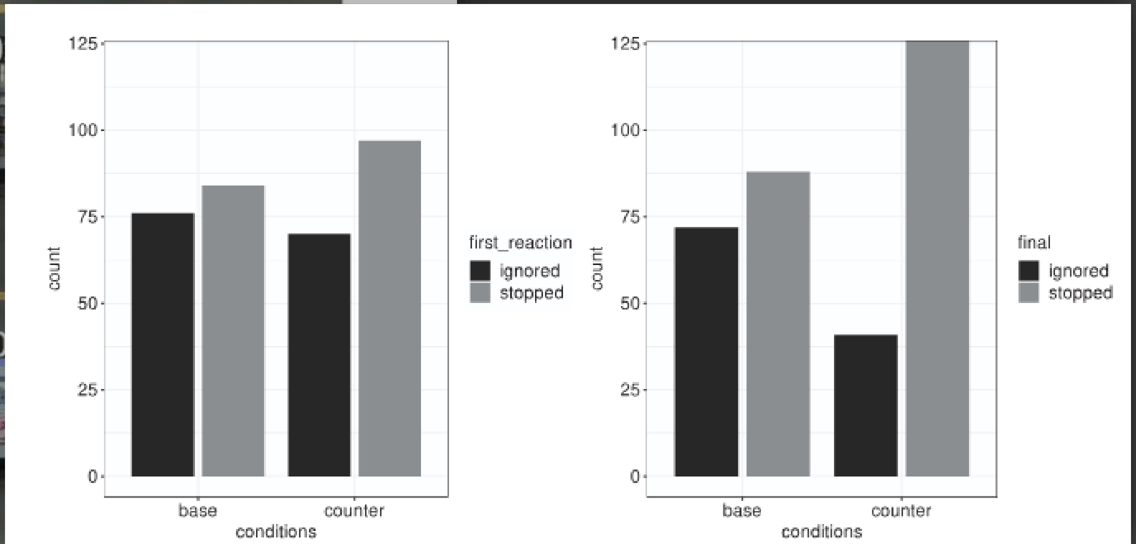
"You might think I am just a robot, but I work for the safety of all of us"

Schneider et al., Stop Ignoring Me! On Fighting the Trivialization of Social Robots in Public Spaces



(c) A pedestrian used his smartphone, then noticed the robot and stopped using it

(d) Two participants that listened to the admonishment and wanted to provoke the robot.



3.6.3 Context/Acceptance/Perceptional

1. National culture and interpersonal trust among humans inversely impacts trust and acceptance of security robots
2. Preferences for human or robot security varies based on role
3. Contextual risk factors significantly influences acceptance

Discussion and Opportunities

- 4.1 Opportunities in Thrust Area 1: Human Factors
- 4.2 Opportunities in Thrust Area 2: Robot Factors
- 4.3 Opportunities in Thrust Area 3: Contextual Factors
- 4.4 Opportunities Across All Thrust Areas

4.1 Opportunities: Human Factors

1. Age
   - More research into acceptance by older demographics
2. Gender
   - Behavioral/emotional response and interface preferences
3. Individual Factors
   - Income, robot experience, interest in science/tech
   - Attitudes toward law enforcement, expectations of security tech

4.2 Opportunities: Robot Factors

1. Physical Appearance
   - Enhancing compliance and affecting emotional response
2. Personalities
   - Designing personality to enhance trust/compliance and reduce stress/fear
3. **Cybersecurity**

4.2.3 Opportunities: Cybersecurity

One reviewed study (Marcu et al.) addressed elevated concerns about security robots being hijacked due to higher capacity for force/weaponry.

Suggested research questions:

1. How does cybersecurity posture influence trust/acceptance?
2. User-friendly authentication mechanisms?
3. Tradeoffs between enhanced security and ease of interaction?
4. Implications of data collection on privacy perceptions?

4.3 Opportunities: Contextual Factors

1. Security Tasks
   - Tasks besides access control, especially unstructured ones
2. Application Domains and Interaction Scenarios
   - Crowded vs. sparse, urban vs. residential, times of day
3. Preference for Humans over Security Robots
   - Further exploration of psychological factors, situations, and empathy
4. Culture Factors
   - Unexplored cultures, trust in institutions, value of privacy
5. Social, Political, and Economic Factors
   - Societal norms, regulatory frameworks, job markets, media portrayals
6. Legal Aspects
   - Privacy, data protection, human rights instruments, AI ethics, policy

4.4 Opportunities: All Thrust Areas

1. Critical or Social Justice Views
   - Authority/public dynamics, unbalanced impact of automated decisions
2. Theoretical Framework
   - Suggestion of comparison of various proposed frameworks
3. Research Settings
   - Indirect images/videos/etc. vs. real robots
4. Behavioral and Perceptional Acceptance Measures
   - Highlighting unmatching behavioral and perceptional acceptance
5. Study Transparency
   - This review highlights mixed results and a lack of reproducibility

Conclusion

- 47 studies reviewed
- 3 thrusts proposed
- 3 outcome areas proposed
- Further research topics identified

- My opinion
  - Human gender, robot design/anthropomorphism, and culture all seem highly important, but also complex and unclear in their impact at this point

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